

Docker



helle

Overview

Topic	Sub-Topics	Duration (in Days)	Duration (in hours)	Total duration: 2 weeks
Introduction and Installation	Docker Introduction, Installation on Various Platforms, Docker CLI Basics	0.5	4	
Containers	Creating and Managing Containers, Container Lifecycle, Docker Run, Exec, Logs	1	8	
Images	Docker Images, Creating and Managing Images, Dockerfile, Image Layers, Best Practices	1	8	
Networking	Docker Networking, Bridge Networks, Host Networks, Overlay Networks, Network Security	1	8	
Docker Compose	Docker Compose Basics, Multi-Container Applications, Volumes, Networks in Compose	1	8	
Registry	Docker Hub, Private Registries, Pushing and Pulling Images, Managing Registries	0.5	4	
Assignment + Level Up		5	40	

Introduction

This document is designed to help you gain practical experience with Docker. By completing these topics, you'll understand how to manage containerized applications and their environments efficiently.

Prerequisites

Docker 20.10 or later, Docker compose 1.29 or later, Basic understanding of Linux command-line operations, Basic knowledge of web application development.

Usage

Create Dockerfiles, define and run multi-container applications with Docker Compose, set up custom networks, and use Docker registries.

Learning Expectations

You will gain practical experience in Docker, understanding Docker Compose and networking, and skills in using Docker registries.

Library Version

Docker 20.10+, Docker Compose 1.29+

Course Material

Topic	Documentation
Introduction and Installation	https://www.tutorialspoint.com/docker/index.htm https://docs.docker.com/guides/getting-started/get-docker-desktop/
Containers	https://docs.docker.com/guides/docker-concepts/the-basics/what-is-a-container/
Images	https://docs.docker.com/guides/docker-concepts/the-basics/what-is-an-image/
Networking	https://docs.docker.com/network/
Docker Compose	https://docs.docker.com/compose/gettingstarted/
Registry	https://docs.docker.com/guides/docker-concepts/the-basics/what-is-a-registry/

Learning modules

Additional Resources

- <https://www.youtube.com/watch?v=iqqDU2crIEQ>
- <https://www.youtube.com/watch?v=SXwC9fSwct8>
- <https://www.youtube.com/watch?v=pg19Z8LL06w>
- <https://www.youtube.com/watch?v=bKFMS5C4CG0>
- <https://www.youtube.com/watch?v=pTFZFxd4hOI&t=9s>
- [Docker Tutorial for Beginners: Basics, Architecture, Containers](#)
- [Docker Tutorial - GeeksforGeeks](#)

Level Up

Docker

<https://nagarro.sharepoint.com/sites/university/SitePages/Pathways.aspx?pathways=Dockers%20fundamentals>

Additional Reference Link:

LinkedIn Learning

- <https://www.linkedin.com/learning/docker-for-developers-14493163/docker-for-developers>
- <https://www.linkedin.com/learning/learning-docker-17236240/get-up-and-running-quick-with-docker>
- <https://www.linkedin.com/learning/docker-essential-training/diving-deeper-into-docker>
- <https://www.linkedin.com/learning/paths/docker-foundations-professional-certificate>